

ALMOST EVERYWHERE MONOTONE ADDITIVE FUNCTIONS

QIYUAN GU

ABSTRACT. We prove that a real additive function which decreases on a set of natural density zero is a nonnegative constant multiple of the logarithm. This answers a conjecture of Erdős from 1946, listed as Problem 1122 on the Erdős Problems website. The main step is to deduce finite concentration without a growth assumption on the values at prime powers. A truncated normalization and a bounded clipping reduce this step to Mangerel’s first-moment theorem for short intervals, Ruzsa’s second-moment estimate, and Elliott’s high-power Turán–Kubilius inequality.

1. INTRODUCTION

A function $f: \mathbb{N} \rightarrow \mathbb{R}$ is *additive* if $f(ab) = f(a) + f(b)$ whenever $(a, b) = 1$. Erdős proved that an additive function which is nondecreasing on $\mathbb{N}$ must be a constant multiple of $\log n$. He asked whether the same conclusion follows when the set of decreases has natural density zero [1, p. 3]. The question is also recorded as Erdős Problem 1122 [10].

Theorem 1.1. *Let $f: \mathbb{N} \rightarrow \mathbb{R}$ be additive. If*

$$D_f(X) := \#\{n \leq X : f(n+1) < f(n)\} = o(X),$$

then there is a constant $c \geq 0$ such that $f(n) = c \log n$ for every $n \in \mathbb{N}$.

Erdős also proved that $f(n+1) - f(n) = o(1)$ implies $f(n) = c \log n$. Kátai and independently Wirsing weakened this to the averaged condition [5, 9]:

$$\frac{1}{X} \sum_{n \leq X} |f(n+1) - f(n)| \longrightarrow 0.$$

Hildebrand later proved Erdős’s density-one version of this assertion [4], while Elliott developed broader quantitative and linear-form variants [3, Ch. 11].

Mangerel established an approximate logarithmic representation under the same density hypothesis [6, Theorem 1.9]. For completely additive functions, he obtained the exact conclusion with additional assumptions on the values at primes and the number of decreases [6, Corollary 1.7]. His short-interval theorem is the analytic input used here.

We say that f has *finite concentration* if there are constants $d > 0$ and $R < \infty$ such that, for arbitrarily large X , some interval of length R contains at least dX of the values $f(n)$, $n \leq X$. The interval is allowed to depend on X . Two results of Erdős reduce Theorem 1.1 to finite concentration. Theorem V of [1] states that finite concentration is equivalent to the existence of $c \in \mathbb{R}$ for which

$$(1.1) \quad \sum_p \frac{\min\{(f(p) - c \log p)^2, 1\}}{p} < \infty.$$

Theorem X of the same paper implies that, under (1.1), the limiting distribution of $f(n+1) - f(n)$ has no mass on $(-\infty, 0)$ only when $f(n) = c \log n$ for all n . Both results apply to general additive

functions, with no restriction on their values at higher prime powers. Theorem X is stated without proof in [1, p. 17]; the source of this final distribution implication is discussed below.

It therefore suffices to prove finite concentration. Assuming the contrary, we normalize f and subtract a logarithmic term so that the truncated quadratic mass of its prime values is fixed. After clipping, the Ruzsa–Elliott moment estimates give a positive variance for a strongly additive approximation, while the density hypothesis and Mangerel’s short-interval theorem force the same variance to satisfy an incompatible upper bound.

Notation. Letters p, q denote primes. We write

$$\mathbb{E}_X F = \frac{1}{X} \sum_{n \leq X} F(n), \quad \mathcal{A}_H F(n) = \frac{1}{H} \sum_{j=0}^{H-1} F(n-j) \quad (n \geq H).$$

Restricted averages, such as $\mathbb{E}_X(F \mathbf{1}_{n \geq H})$, retain the normalizing factor $1/X$. For a strongly additive function $z(n) = \sum_{p|n} a_p$, put

$$\mu_z(X) = \sum_{p \leq X} \frac{a_p}{p}.$$

All implicit constants are absolute unless a dependence is indicated. The constants K and M used in the proof are fixed before X tends to infinity. In statements about a family z_X , the prime coefficients may depend on X .

2. MOMENT ESTIMATES AND SHORT INTERVALS

For an additive function z , define

$$A_0(z, X) = \sum_{p^k \leq X} \frac{z(p^k)}{p^k} \left(1 - \frac{1}{p}\right), \quad \tilde{A}_0(z, X) = \sum_{p^k \leq X} \frac{z(p^k)}{p^k},$$

and

$$B_j(z, X)^j = \sum_{p^k \leq X} \frac{|z(p^k)|^j}{p^k} \quad (j = 2, 4).$$

Ruzsa’s second-moment estimate [7], in the form recorded by Mangerel [6, Lemma 3.3], gives

$$(2.1) \quad \mathbb{E}_X |z - A_0(z, X)|^2 \asymp \inf_{b \in \mathbb{R}} \left(b^2 + \sum_{p^k \leq X} \frac{|z(p^k) - b \log(p^k)|^2}{p^k} \right).$$

Elliott’s high-power Turán–Kubilius inequality [2, Theorem 1], with exponent 4, gives

$$\mathbb{E}_X |z - \tilde{A}_0(z, X)|^4 \ll B_2(z, X)^4 + B_4(z, X)^4.$$

Since

$$|\tilde{A}_0(z, X) - A_0(z, X)| \leq \sum_{p^k \leq X} \frac{|z(p^k)|}{p^{k+1}} \ll B_2(z, X),$$

the same bound holds with A_0 as center:

$$(2.2) \quad \mathbb{E}_X |z - A_0(z, X)|^4 \ll B_2(z, X)^4 + B_4(z, X)^4.$$

If z is strongly additive, then

$$(2.3) \quad \begin{aligned} |\tilde{A}_0(z, X) - \mu_z(X)| &\leq \sum_{p \leq X} \frac{|a_p|}{p(p-1)}, \\ |\tilde{A}_0(z, X) - A_0(z, X)| &\leq \sum_{p^k \leq X} \frac{|a_p|}{p^{k+1}}. \end{aligned}$$

Both quantities are $\ll (\sum_{p \leq X} a_p^2/p)^{1/2}$. Also $B_2(z, X)^2 \leq 2 \sum_{p \leq X} a_p^2/p$, with the analogous bound for B_4 . Thus, if $|a_p| \leq L$ and $\sum_{p \leq X} a_p^2/p \leq V$, then

$$(2.4) \quad \mathbb{E}_X |z - \mu_z(X)|^2 \ll V, \quad \mathbb{E}_X |z - \mu_z(X)|^4 \ll V^2 + L^2 V.$$

If the coefficients are bounded and $a_p \rightarrow 0$ for each fixed prime, dominated convergence in (2.3) gives $A_0(z, X) - \mu_z(X) \rightarrow 0$.

Lemma 2.1. *Let z_X be strongly additive functions satisfying*

$$|z_X(p)| \leq L, \quad \sum_{p \leq X} \frac{z_X(p)^2}{p} \leq V$$

for fixed $L, V < \infty$. Then

$$(2.5) \quad \lim_{H \rightarrow \infty} \limsup_{X \rightarrow \infty} \frac{1}{X} \sum_{H \leq n \leq X} |\mathcal{A}_H z_X(n) - \mu_{z_X}(X)|^2 = 0,$$

where H runs through the positive integers.

Proof. For $Y \leq X$, write

$$d_Y = \frac{2}{Y} \sum_{Y/2 < n \leq Y} z_X(n), \quad F_Y(n) = \mathcal{A}_H z_X(n) - d_Y.$$

Mangerel's Theorem 1.1 [6] gives, for every integer $10 \leq H \leq Y/100$,

$$(2.6) \quad \frac{2}{Y} \sum_{Y/2 < n \leq Y} |F_Y(n)| \ll \left(\sqrt{\frac{\log \log H}{\log H}} + (\log Y)^{-1/800} \right) \sqrt{2V}.$$

Here $B_2(z_X, Y)^2 \leq 2V$, and the estimate is uniform over additive functions, so it applies to the family z_X .

Counting multiples of each prime gives

$$(2.7) \quad d_Y - \mu_{z_X}(Y) = \sum_{p \leq Y} z_X(p) \left(\frac{2}{Y} \left(\left\lfloor \frac{Y}{p} \right\rfloor - \left\lfloor \frac{Y}{2p} \right\rfloor \right) - \frac{1}{p} \right) = O\left(\frac{L\pi(Y)}{Y}\right).$$

Jensen's inequality and (2.4) at Y imply

$$(2.8) \quad \frac{2}{Y} \sum_{Y/2 < n \leq Y} |F_Y(n)|^4 \ll V^2 + L^2 V + O_L((\log Y)^{-4}).$$

To see this, center the backward average at $\mu_{z_X}(Y)$. Every index in its inner sums lies in $[1, Y]$ and occurs at most H times. The remaining constant is estimated by (2.7). Hölder's inequality now gives

$$\frac{2}{Y} \sum |F_Y|^2 \leq \left(\frac{2}{Y} \sum |F_Y| \right)^{2/3} \left(\frac{2}{Y} \sum |F_Y|^4 \right)^{1/3}.$$

Together with (2.6) and (2.8), this proves the required iterated limit on each fixed dyadic band $Y \asymp X$.

This dyadic conclusion extends to the whole interval. On each of finitely many bands with $Y \asymp X$, we have

$$|\mu_{z_X}(X) - \mu_{z_X}(Y)| \leq L \sum_{Y < p \leq X} \frac{1}{p} = o(1).$$

The omitted initial segment $H \leq n \leq \eta X$ contributes $O_{L,V}(\sqrt{\eta})$ to (2.5): apply Cauchy–Schwarz and Jensen, using the fourth-moment bound (2.4) at X . First take $X \rightarrow \infty$, then $H \rightarrow \infty$, and finally $\eta \downarrow 0$. $\square$

3. A WEIGHTED ESTIMATE FOR A SET OF PRIMES

Lemma 3.1. *Suppose $0 < M < 1/4$ and $\mathcal{T}$ is a set of primes at most X such that $\sum_{p \in \mathcal{T}} 1/p \leq M$. Then, uniformly in $\mathcal{T}$,*

$$(3.1) \quad \sum_{p \in \mathcal{T}} \frac{1}{p} \sum_{X/p < q \leq X} \frac{1}{q} \ll M \log(2/M) + o_{X \rightarrow \infty}(1).$$

Here M is fixed in the term $o(1)$.

Proof. Write $P(y) = \sum_{p \leq y} 1/p$. Chebyshev's bound gives $\vartheta(t) := \sum_{p \leq t} \log p \leq Dt$ with $D = \log 4$; see [8]. Abel summation therefore gives, for $2 \leq y \leq z$,

$$\begin{aligned} P(z) - P(y) &= \frac{\vartheta(z)}{z \log z} - \frac{\vartheta(y)}{y \log y} + \int_y^z \frac{\vartheta(t)(\log t + 1)}{t^2(\log t)^2} dt \\ &\leq D \left(\log \frac{\log z}{\log y} + \frac{1}{\log y} \right). \end{aligned}$$

Take X large enough that $X^M \geq 2$. Interchanging the sums, the expression in (3.1) is

$$\sum_{q \leq X} \frac{1}{q} \sum_{\substack{p \in \mathcal{T} \\ X/q < p}} \frac{1}{p}.$$

For $q > X^M$, bound the inner sum by M . This part is at most

$$M(P(X) - P(X^M)) \leq DM \log(1/M) + \frac{D}{\log X}.$$

For $2 \leq q \leq X^M$, put $v = \log q / \log X \leq M < 1/4$ and discard the restriction $p \in \mathcal{T}$. Since $\log(X/q) \geq \frac{1}{2} \log X$ and $X/q \geq 2$, the same estimate gives

$$\begin{aligned} P(X) - P(X/q) &\leq D \left(-\log(1-v) + \frac{2}{\log X} \right) \\ &\leq D \left(2 + \frac{2}{\log 2} \right) \frac{\log q}{\log X}. \end{aligned}$$

Here we used $-\log(1-v) \leq 2v$ and $\log q \geq \log 2$. Partial summation also gives

$$\sum_{q \leq y} \frac{\log q}{q} = \frac{\vartheta(y)}{y} + \int_2^y \frac{\vartheta(t)}{t^2} dt \leq D(1 + \log y).$$

Thus the part with $q \leq X^M$ is at most $D^2(2 + 2/\log 2)(1 + 1/\log 2)M$. Combining the two parts proves

$$\sum_{p \in \mathcal{T}} \frac{1}{p} \sum_{X/p < q \leq X} \frac{1}{q} \leq CM \log(2/M) + \frac{D}{\log X}, \quad C = D + \frac{(2 + 2/\log 2)D^2(1 + 1/\log 2)}{\log 2}.$$

In particular, C is independent of M , X , and $\mathcal{T}$. $\square$

4. NORMALIZATION

In the remainder of the proof, suppose that f does not have finite concentration. For $s > 0$ define

$$(4.1) \quad W_X(s) = \min_{c \in \mathbb{R}} \left\{ c^2 + \sum_{p \leq X} \frac{\min\{(f(p)/s - c \log p)^2, 1\}}{p} \right\}.$$

Lemma 4.1. *For each fixed $M > 0$ and all sufficiently large X , there are $s_X > 0$ and a minimizer c_X in (4.1) such that $W_X(s_X) = M$. They may be chosen so that $s_X \rightarrow \infty$; any such choices satisfy $c_X \rightarrow 0$. If*

$$u_p = \frac{f(p)}{s_X} - c_X \log p \quad (p \leq X),$$

then

$$(4.2) \quad \sum_{p \leq X} \frac{\min\{u_p^2, 1\}}{p} = M - c_X^2 = M - o(1),$$

$$(4.3) \quad \sum_{\substack{p \leq X \\ |u_p| < 1}} \frac{u_p \log p}{p} = c_X.$$

No u_p has absolute value exactly 1.

Proof. The expression in braces in (4.1) is continuous and coercive as a function of c , so its minimum is attained. Every minimizer satisfies $c^2 \leq \sum_{p \leq X} 1/p$, by comparison with $c = 0$. The minimum can therefore be taken over a fixed compact interval, independent of s , and is continuous in $s > 0$. On writing $\lambda = sc$, the expression becomes

$$\frac{\lambda^2}{s^2} + \sum_{p \leq X} \frac{\min\{(f(p) - \lambda \log p)^2/s^2, 1\}}{p}.$$

It follows that $W_X(s)$ is nonincreasing in s , and it tends to zero as $s \rightarrow \infty$ for fixed X .

For each fixed $s > 0$, $W_X(s) \rightarrow \infty$. Otherwise some unbounded sequence of X would have bounded $W_X(s)$ and bounded minimizers. Pass to a subsequence along which the minimizers converge to c . Fatou's lemma gives

$$\sum_p \frac{\min\{(f(p)/s - c \log p)^2, 1\}}{p} < \infty.$$

Rescaling by the fixed number s preserves convergence of the truncated square sum. Erdős's Theorem V would therefore give finite concentration of f , contrary to the assumption.

Continuity now supplies s_X with $W_X(s_X) = M$. Monotonicity and the preceding claim imply $s_X \rightarrow \infty$. We have $|c_X| \leq \sqrt{M}$. If a subsequential limit of c_X were a nonzero number c , Fatou's lemma on the fixed primes would give

$$\sum_p \frac{\min\{(c \log p)^2, 1\}}{p} \leq M,$$

which is impossible. Thus $c_X \rightarrow 0$, and (4.2) follows.

It remains to prove the stationary identity and exclude equality at the truncation threshold. Let $G(c)$ denote the expression in braces in (4.1), with $s = s_X$. For any subset J of the primes

at most X , the quadratic

$$Q_J(c) = c^2 + \sum_{p \in J} \frac{(f(p)/s_X - c \log p)^2}{p} + \sum_{\substack{p \leq X \\ p \notin J}} \frac{1}{p}$$

lies above $G(c)$. Choose $J = \{p \leq X : |u_p| < 1\}$. Then $Q_J(c_X) = G(c_X)$, so c_X also minimizes Q_J . The equation $Q'_J(c_X) = 0$ is exactly (4.3). If $|u_p| = 1$ for some p , the quadratic $Q_{J \cup \{p\}}$ also touches G at c_X and must have derivative zero there, contradicting

$$Q'_{J \cup \{p\}}(c_X) - Q'_J(c_X) = -\frac{2u_p \log p}{p} \neq 0. \quad \square$$

5. CLIPPING THE NORMALIZED FUNCTION

Fix $K > 1$ and $0 < M < 1/4$, and use Lemma 4.1. Let $\phi_K(t) = \max\{-K, \min\{t, K\}\}$ and set

$$\begin{aligned} \mathcal{T} &= \{p \leq X : |u_p| > 1\}, & S_0(n) &= \sum_{\substack{p|n \\ p \notin \mathcal{T}}} u_p, \\ a_X &= \sum_{\substack{p \leq X \\ p \notin \mathcal{T}}} \frac{u_p}{p}, & S(n) &= S_0(n) - a_X. \end{aligned}$$

Coefficients at primes greater than X are set to zero. Define

$$\begin{aligned} z_X(n) &= \sum_{p|n} \phi_K(u_p), & Z(n) &= z_X(n) - a_X, \\ (5.1) \quad Y(n) &= \phi_K\left(\frac{f(n)}{s_X} - c_X \log n - a_X\right). \end{aligned}$$

Thus Y clips the value of the additive function, whereas z_X clips the prime contributions individually.

Lemma 5.1. *There is an absolute constant C such that*

$$(5.2) \quad \limsup_{X \rightarrow \infty} \mathbb{E}_X |Y - Z|^2 \leq C \left(\frac{M}{K^2} + M^2 \log(2/M) + K^2 M^2 \right).$$

Proof. The small coefficients have quadratic mass at most M , and $\sum_{p \in \mathcal{T}} 1/p \leq M$. Hence

$$(5.3) \quad \mathbb{E}_X S^4 \ll M + M^2.$$

Put $T(n) = \#\{p \in \mathcal{T} : p | n\}$. Counting multiples of distinct primes gives

$$(5.4) \quad \mathbb{E}_X T(T-1) = \sum_{\substack{p, q \in \mathcal{T} \\ p \neq q}} \frac{1}{X} \left\lfloor \frac{X}{pq} \right\rfloor \leq M^2.$$

Moreover,

$$(5.5) \quad \mathbb{E}_X (S^2 \mathbf{1}_{T \geq 1}) \ll M^2 \log(2/M) + o(1).$$

For $p \in \mathcal{T}$, the coefficient of p in S_0 is zero, so $S_0(pm) = S_0(m)$ for every m , including multiples of p . Writing $y = X/p$, for $y \geq 2$ equation (2.4) gives

$$\begin{aligned} \mathbb{E}_X (S^2 \mathbf{1}_{p|n}) &= \frac{1}{p} \frac{1}{y} \sum_{m \leq y} |S_0(m) - a_X|^2 \\ &\ll \frac{1}{p} \left(M + |a_X - \mu_{S_0}(y)|^2 \right). \end{aligned}$$

For $1 \leq y < 2$, the same bound follows directly from $S_0(1) = \mu_{S_0}(y) = 0$. By Cauchy–Schwarz,

$$|a_X - \mu_{S_0}(X/p)|^2 \leq M \sum_{X/p < q \leq X} \frac{1}{q}.$$

Hence, using $\mathbf{1}_{T \geq 1} \leq \sum_{p \in \mathcal{T}} \mathbf{1}_{p|n}$,

$$\begin{aligned} \mathbb{E}_X(S^2 \mathbf{1}_{T \geq 1}) &\ll M \sum_{p \in \mathcal{T}} \frac{1}{p} + M \sum_{p \in \mathcal{T}} \frac{1}{p} \sum_{X/p < q \leq X} \frac{1}{q} \\ &\ll M^2 + M^2 \log(2/M) + o(1), \end{aligned}$$

by Lemma 3.1. This proves (5.5).

Set

$$Y_*(n) = \phi_K \left(S(n) + \sum_{\substack{p|n \\ p \in \mathcal{T}}} u_p \right).$$

If $T(n) = 0$, the difference $Y_*(n) - Z(n)$ vanishes for $|S(n)| \leq K$ and otherwise has absolute value at most $|S(n)|$. Its squared contribution is at most $\mathbb{E}_X S^4 / K^2$. If $T(n) = 1$, with unique tail value v , the Lipschitz property of ϕ_K gives

$$|\phi_K(S + v) - S - \phi_K(v)| \leq 2|S|.$$

If $T(n) \geq 2$, then

$$|Y_*(n) - Z(n)| \leq |S(n)| + K(T(n) + 1) \leq |S(n)| + \frac{3}{2}KT(n),$$

and hence

$$|Y_*(n) - Z(n)|^2 \leq CS(n)^2 + CK^2T(n)(T(n) - 1).$$

Equations (5.3)–(5.5) therefore imply

$$(5.6) \quad \mathbb{E}_X |Y_* - Z|^2 \ll \frac{M}{K^2} + M^2 \log(2/M) + K^2 M^2 + o(1).$$

To compare Y and Y_* , suppose $p^k || n$ with $k \geq 2$. Its contribution to the difference between the two unclipped expressions is

$$\frac{f(p^k) - f(p)}{s_X} - c_X(k-1) \log p.$$

This tends to zero for each fixed prime power. Moreover, $\sum_{p, k \geq 2} p^{-k} < \infty$. Given $\varepsilon > 0$, choose a finite set $\mathcal{F}$ of prime powers such that

$$\sum_{\substack{p^k \notin \mathcal{F} \\ k \geq 2}} p^{-k} < \varepsilon.$$

Then, uniformly in X ,

$$\frac{1}{X} \#\{n \leq X : \text{some } p^k \notin \mathcal{F}, k \geq 2, \text{ divides } n\} < \varepsilon.$$

Outside this exceptional set, only the fixed prime powers in $\mathcal{F}$ contribute, and their total contribution tends uniformly to zero. Thus the unclipped difference tends to zero in probability. The function ϕ_K is bounded and Lipschitz, so

$$\mathbb{E}_X |Y - Y_*|^2 = o(1).$$

Together with (5.6), this proves the lemma. $\square$

Lemma 5.2. *There are absolute constants $c_0, C_0 > 0$ such that*

$$(5.7) \quad \liminf_{X \rightarrow \infty} \mathbb{E}_X |z_X - \mu_{z_X}(X)|^2 \geq c_0 M - C_0 K^2 M^2.$$

Proof. Write $v_p = \phi_K(u_p)$. Since $K > 1$, (4.2) implies

$$(5.8) \quad M - o(1) \leq \sum_{p \leq X} \frac{v_p^2}{p} \leq K^2 M.$$

On $p \notin \mathcal{T}$ we have $v_p = u_p$, so (4.3) gives

$$(5.9) \quad \left| \sum_{p \leq X} \frac{v_p \log p}{p} \right| \leq |c_X| + KM \log X.$$

Only the lower bound $\sum_{p \leq X} (\log p)^2 / p \gg (\log X)^2$ is needed here. Indeed, Chebyshev's bound $\vartheta(t) \gg t$, followed by partial summation with $\log t/t$, gives this bound from $\int^X (\log t - 1) dt/t \asymp (\log X)^2$. The quadratic projection identity therefore gives

$$(5.10) \quad \begin{aligned} \inf_{b \in \mathbb{R}} \sum_{p \leq X} \frac{(v_p - b \log p)^2}{p} &= \sum_{p \leq X} \frac{v_p^2}{p} - \frac{(\sum_{p \leq X} v_p \log p / p)^2}{\sum_{p \leq X} (\log p)^2 / p} \\ &\geq M - CK^2 M^2 - o(1). \end{aligned}$$

Apply (2.1), dropping the nonnegative b^2 term and the terms with $k \geq 2$ from its right-hand side. For each fixed prime p , $v_p \rightarrow 0$, because $s_X \rightarrow \infty$ and $c_X \rightarrow 0$. Therefore (2.3) and dominated convergence give $A_0(z_X, X) - \mu_{z_X}(X) \rightarrow 0$. The second moments are uniformly bounded by (2.4) and (5.8); changing the center consequently changes the second moment by $o(1)$. Combining this with (5.10) proves (5.7). $\square$

6. PROOF OF THEOREM 1.1

Assume for contradiction that f has no finite concentration. The function Y in (5.1) is bounded between $-K$ and K . If $f(n+1) \geq f(n)$, monotonicity and Lipschitz continuity of ϕ_K imply

$$(Y(n) - Y(n+1))_+ \leq |c_X| \log(1 + 1/n),$$

where $t_+ = \max\{t, 0\}$. At a decrease of f , the same quantity is at most $2K$. Telescoping the signed increments gives

$$(6.1) \quad \sum_{n < X} |Y(n+1) - Y(n)| \leq 2K + 4KD_f(X) + 2|c_X| \log X = o(X).$$

For each fixed H , every difference $Y(n) - Y(n-j)$ is bounded by the sum of the j intervening absolute increments. Since $|Y| \leq K$, (6.1) yields

$$(6.2) \quad \frac{1}{X} \sum_{H \leq n \leq X} |Y(n) - \mathcal{A}_H Y(n)|^2 \rightarrow 0.$$

Let $m_X = \mu_{z_X}(X) - a_X$. Then $Z - m_X = z_X - \mu_{z_X}(X)$, and

$$(6.3) \quad Z - m_X = (Z - Y) + (Y - \mathcal{A}_H Y) + \mathcal{A}_H(Y - Z) + (\mathcal{A}_H Z - m_X).$$

The square of the right-hand side is at most four times the sum of the four squared terms. Jensen's inequality gives the contraction

$$\frac{1}{X} \sum_{H \leq n \leq X} |\mathcal{A}_H(Y - Z)(n)|^2 \leq \frac{1}{X} \sum_{n \leq X} |Y(n) - Z(n)|^2.$$

The prime coefficients of z_X are bounded by K and have quadratic mass at most K^2M , so Lemma 2.1 applies to the last term of (6.3). The first H indices contribute $o(1)$ to the centered second moment: this follows directly from (2.4) and Cauchy–Schwarz. Taking $X \rightarrow \infty$ and then $H \rightarrow \infty$ in (6.3), and using (6.2), we obtain

$$(6.4) \quad \begin{aligned} \limsup_X \mathbb{E}_X |z_X - \mu_{z_X}(X)|^2 &\leq 8 \limsup_X \mathbb{E}_X |Y - Z|^2 \\ &\leq C_1 \left(\frac{M}{K^2} + M^2 \log(2/M) + K^2 M^2 \right) \end{aligned}$$

for an absolute constant C_1 .

Choose $K > 1$ so large that $C_1/K^2 < c_0/4$, where c_0 is from Lemma 5.2. Next choose $0 < M < 1/4$ so small that

$$C_1 M \log(2/M) + (C_1 + C_0) K^2 M < c_0/4.$$

With these fixed choices, (6.4) contradicts (5.7). Hence f has finite concentration.

Erdős’s Theorem V now gives a constant c for which the additive remainder $r(n) = f(n) - c \log n$ satisfies

$$\sum_p \frac{\min\{r(p)^2, 1\}}{p} < \infty.$$

Erdős’s Theorem X gives a limiting distribution for $f(n+1) - f(n)$, and absence of mass on $(-\infty, 0)$ forces r to be identically zero. For each $\varepsilon > 0$, the hypothesis gives

$$\frac{1}{X} \#\{n \leq X : f(n+1) - f(n) < -\varepsilon\} \leq \frac{D_f(X)}{X} \rightarrow 0.$$

The limiting distribution therefore has no mass on $(-\infty, 0)$. It follows that $f(n) = c \log n$ for all n . Finally, $c < 0$ would give a decrease at every integer, so $c \geq 0$. $\square$

Remark 6.1. All estimates for prime sums used above follow from Chebyshev’s bounds; neither Mertens’s theorem nor the prime number theorem is needed.

Remark 6.2. Erdős states Theorems IX and X without proof on page 17 of [1]; Theorem XI subsequently uses Theorem X. A later source for the distribution argument is Hildebrand’s Erdős–Wintner theorem for differences [4]. Its limiting distribution is symmetric, so absence of mass on $(-\infty, 0)$ makes it a point mass at zero. Hildebrand’s density-one characterization of the logarithm then gives the required conclusion.

DECLARATION OF GENERATIVE AI USE

GPT-6 Astra was used to generate the mathematical proofs and draft the manuscript. GPT-5.6 Sol and Claude Opus 5 were used only for editorial review of the exposition. The author reviewed the final manuscript and takes full responsibility for its content. The accompanying Lean 4 formalization checks the implication from the five cited results, which remain explicit hypotheses; it does not formalize their proofs. The Lean code was generated with OpenAI Codex (GPT-6).

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UNIVERSITY OF CHICAGO

Email address: phoenix1203@uchicago.edu